

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO4: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Mock Interview

Duration: 1hr

1. A machine learning engineer applies Reinforcement Learning to solve sequential decision-making problems. Explain how an agent learns optimal behavior through interaction with the environment. (5 Marks)

Reinforcement Learning (RL) is a machine learning approach in which an agent learns by interacting with an environment. At each step, the agent observes the current state and selects an action. The environment then provides a reward and moves to a new state.

The agent tries to maximize the cumulative reward over time. Initially, it explores different actions. Based on the rewards received, it updates its policy or value function. Actions that produce higher long-term rewards become more likely to be selected. Through repeated interaction, the agent learns an optimal policy that gives the best action for each state.

Main components: Agent → State → Action → Environment → Reward → New State.

2. If you are asked to design a Reinforcement Learning solution for a real-time system such as ride-sharing or delivery optimization, explain how you would formulate the problem using a Markov Decision Process. (5 Marks)

A real-time ride-sharing or delivery optimization problem can be formulated as a Markov Decision Process (MDP) using the following components:

- **State (S):** Current location of vehicles, available drivers, pending requests, traffic conditions, fuel/battery level, and delivery status.
- **Action (A):** Assign a driver, select a route, accept/reject a request, or change the delivery sequence.
- **Transition (P):** Represents how the system changes after an action, such as a vehicle reaching a new location or a new customer request arriving.
- **Reward (R):** Positive reward for completing rides/deliveries efficiently and negative reward for delays, high fuel consumption, cancellations, or increased travel distance.
- **Discount factor (γ):** Determines the importance of future rewards.

The RL agent learns a policy that maximizes the total expected reward while continuously adapting to changing traffic and demand.

3. Compare model-free and model-based Reinforcement Learning approaches. Explain their working principles and how they handle uncertainty. (5 Marks)

Answer:

Feature	Model-Free RL	Model-Based RL
Environment model	Does not require a model	Learns or uses a model
Learning	Learns directly from experience	Learns how the environment behaves
Planning	Usually limited	Can plan future actions
Data efficiency	Generally requires more experience	Usually more data-efficient
Examples	Q-Learning, SARSA, DQN	Dyna-Q, Model Predictive Control with learned models

Model-Free RL directly learns the value of actions or a policy from rewards. It does not try to predict the next state.

Model-Based RL learns or uses a model of state transitions and rewards. It can simulate possible future outcomes before selecting an action.

Regarding **uncertainty**, model-free methods handle it mainly through repeated experience and exploration, while model-based methods can represent uncertainty in their learned environment model and use it during planning.

4. A recommendation system using Reinforcement Learning is underperforming. Explain how reward sparsity and hyperparameter tuning can be diagnosed and improved. (5 Marks)

If an RL recommendation system is underperforming, two important areas to investigate are reward sparsity and hyperparameter settings.

Reward Sparsity

Reward sparsity occurs when the agent receives rewards only occasionally. This makes it difficult to determine which actions caused the reward.

It can be improved by:

- Designing more informative intermediate rewards.
- Using reward shaping.

- Providing feedback for useful user interactions such as clicks, watch time, or successful recommendations.
- Using techniques such as experience replay and multi-step learning.

Hyperparameter Tuning

Important parameters include:

- Learning rate
- Discount factor (γ)
- Exploration rate (ϵ)
- Batch size
- Replay buffer size

Different values can be tested using validation environments or simulations. Performance should be compared using metrics such as cumulative reward, click-through rate, and user satisfaction.

5. If you are implementing an RL-based game-playing AI, describe how you would design the learning pipeline and ensure training stability. (5 Marks)

The learning pipeline for an RL-based game AI can be designed as:

Game Environment → State Observation → Agent → Action → Reward → Experience → Learning → Updated Policy

First, the game environment provides the current state. The agent selects an action using its policy. The game returns a reward and a new state. These experiences are stored and used to update the agent.

To ensure training stability:

- Use experience replay to reuse past experiences.
- Use a target network for stable value estimation in DQN.
- Normalize rewards when appropriate.
- Use an appropriate learning rate.
- Balance exploration and exploitation using ϵ -greedy or similar strategies.
- Monitor training using cumulative reward, win rate, and loss.
- Test the trained agent separately from the training environment.

6. A company wants to use Reinforcement Learning for dynamic pricing. Explain how the agent learns from interaction and improves pricing decisions over time. (5 Marks)

In dynamic pricing, the RL agent decides the price of a product or service based on current market conditions.

The **state** may include customer demand, inventory, time, competitor prices, and historical sales. The **action** is the price selected by the agent. The **reward** can be the profit or revenue generated after applying that price.

For example, if a higher price increases profit without significantly reducing demand, the agent receives a positive reward. If the price is too high and demand decreases, the reward becomes lower.

By repeatedly interacting with the market or a simulated environment, the agent updates its policy and gradually learns which prices produce higher long-term profits.

7. If an RL model is not converging properly, explain how you would identify and fix issues related to learning rate and exploration strategy. (5 Marks)

If an RL model is not converging, the **learning rate** and **exploration strategy** should be examined.

Learning Rate

A very high learning rate can cause unstable updates and make the model overshoot good solutions. A very low learning rate can make learning extremely slow.

Possible solutions:

- Reduce the learning rate if training is unstable.
- Increase it slightly if learning is too slow.
- Use learning-rate decay or adaptive optimizers where appropriate.

Exploration Strategy

If exploration is too high, the agent keeps selecting random actions and does not exploit learned knowledge. If exploration is too low, the agent may get stuck in a poor policy.

A common solution is ϵ -greedy exploration, where ϵ starts high and gradually decreases.

Training curves, reward trends, loss values, and policy behavior should be monitored to identify the problem.

8. A robotics system uses Reinforcement Learning for navigation. Explain how the environment and agent interaction leads to optimal path learning. (5 Marks)

In an RL-based navigation system, the robot is the agent and its surroundings represent the environment.

The robot observes its state, which may include its position, distance from obstacles, sensor readings, and destination location. It then selects an action, such as moving forward, backward, left, or right.

The environment provides a reward:

- Positive reward for moving toward the destination.
- Large positive reward for reaching the target.
- Negative reward for collisions.
- Small penalties for unnecessary movement or long paths.

The robot learns through repeated trials. Over time, it identifies actions that produce higher cumulative rewards and learns a policy that leads to shorter, safer, and more efficient paths.

9. If you are asked to improve an RL model with delayed rewards, explain techniques to handle credit assignment problems. (5 Marks)

In many RL problems, an action may produce a reward only after several steps. This creates a **credit assignment problem**, because the agent has difficulty determining which earlier action was responsible for the final reward.

Techniques to handle this include:

1. **Eligibility traces:** Give credit to recently performed actions according to their relevance.

2. **n-step returns:** Use rewards from multiple future steps when updating values.
3. **Temporal-Difference learning:** Propagate information about future rewards backward through previous states.
4. **Reward shaping:** Add meaningful intermediate rewards to provide more frequent feedback.
5. **Monte Carlo methods:** Use complete episodes to estimate the value of earlier actions.

These techniques help the agent learn which actions contributed to long-term success.

10. A financial firm uses Reinforcement Learning for trading decisions. Explain how uncertainty and risk can be handled in the learning process. (5 Marks)

In financial trading, the RL agent can make decisions such as buy, sell, or hold. The state may contain market prices, technical indicators, portfolio balance, and recent market trends.

Financial markets are highly uncertain, so the reward should not depend only on profit. The system should also consider risk and losses.

Risk can be handled by:

- Including risk-adjusted returns in the reward function.
- Penalizing excessive losses and large portfolio fluctuations.
- Limiting position sizes.
- Using stop-loss or risk constraints.
- Training with different historical market conditions.
- Testing the policy on unseen market data.

The agent learns from market interactions and updates its policy based on both returns and risk. This helps it develop a strategy that aims for higher returns while controlling risk, rather than simply maximizing short-term profit.

Code of Conduct:

I P. Sahith Sai (192425029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Sahith Sai

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately